

Homework solutions for 02YELMA

Chapter: Mathematical tools

1.

$$\begin{aligned}f(x, y, z) &= xyz \\ \nabla f &= (yz, xz, xy) \\ \nabla f(1, 1, 1) &= (1, 1, 1), \quad |\nabla f| = \sqrt{3}\end{aligned}$$

The gradient points in the direction of maximum increase of the function, and its magnitude gives the maximum rate of change. The directional derivative in any unit direction $\hat{u}$ is $\nabla f \cdot \hat{u}$.

2.

$$\begin{aligned}\vec{F} &= (x^2, -y^2, z^2) \\ \nabla \cdot \vec{F} &= 2x - 2y + 2z \\ \nabla \cdot \vec{F}(1, -1, 1) &= 6\end{aligned}$$

A positive divergence indicates that the point behaves like a source, meaning the field locally flows outward from that point.

3.

$$\begin{aligned}\vec{F} &= (y, z, x) \\ \nabla \times \vec{F} &= (-1, -1, -1)\end{aligned}$$

The curl represents local rotation. A constant nonzero curl indicates uniform rotational behavior, with the direction of the curl vector giving the axis of rotation.

4.

$$\nabla \times (\nabla T) = 0$$

This follows from equality of mixed partial derivatives.

$$\nabla \cdot (\nabla \times \vec{T}) = 0$$

This also follows from cancellation of mixed partial derivatives. See Griffiths, Introduction to Electrodynamics, 3th edition, section 1.2.7 for more details.

5. For example, let $\vec{A} = \vec{B}$ and let $\vec{C}$ be perpendicular to both. Then the directions of $(\vec{A} \times \vec{B}) \times \vec{C}$ and $\vec{A} \times (\vec{B} \times \vec{C})$ would not be the same, so the triple product is not associative in general.

6.

$$\begin{aligned}\vec{F} &= (x, y) \\ \oint \vec{F} \cdot d\vec{r} &= 0\end{aligned}$$

The field is conservative since $\vec{F} = \nabla \left(\frac{x^2+y^2}{2} \right)$, so the line integral over any closed path is zero. Or, if you really want to compute it:

$$\vec{F} = (x, y), \quad \oint \vec{F} \cdot d\vec{r} = \int_{AB} + \int_{BC} + \int_{CA}$$

AB: $x = t, y = 1, t \in [0, 1]$

$$\int_{AB} \vec{F} \cdot d\vec{r} = \int_0^1 t \, dt = \frac{1}{2}$$

BC: $x = 1 - t, y = 1 - t, t \in [0, 1]$

$$\int_{BC} \vec{F} \cdot d\vec{r} = \int_0^1 -2(1 - t) \, dt = -1$$

CA: $x = 0, y = t, t \in [0, 1]$

$$\int_{CA} \vec{F} \cdot d\vec{r} = \int_0^1 t \, dt = \frac{1}{2}$$

$$\oint \vec{F} \cdot d\vec{r} = \frac{1}{2} - 1 + \frac{1}{2} = 0$$

$$\vec{F} = \nabla \left(\frac{x^2 + y^2}{2} \right) \Rightarrow \oint \vec{F} \cdot d\vec{r} = 0$$

7.

$$\vec{F} = (x, y, z) = \vec{r}$$

$$\iint \vec{F} \cdot d\vec{A} = 4\pi$$

This gives the total outward flux through the sphere, representing the net strength of a source at the origin.

8.

$$\rho = x + y + z$$

$$M = \int_0^3 \int_0^{2\pi} \int_0^1 (r \cos \theta + r \sin \theta + z) r \, dr \, d\theta \, dz$$

$$M = \frac{9\pi}{2}$$

The terms involving $\cos \theta$ and $\sin \theta$ vanish due to symmetry, leaving only the contribution from z .

9.

$$\vec{F} = (x^2, y^2, z^2)$$

$$\nabla \cdot \vec{F} = 2x + 2y + 2z$$

$$\iiint (2x + 2y + 2z) \, dV = 3$$

The Divergence Theorem converts the surface flux into a volume integral of the divergence.

10.

$$\begin{aligned}\vec{F} &= (-y, x, z) \\ \nabla \times \vec{F} &= (0, 0, 2) \\ \oint \vec{F} \cdot d\vec{r} &= 2\pi\end{aligned}$$

Stokes' Theorem relates circulation around the boundary to the flux of the curl through the surface.

11.

$$\begin{aligned}\vec{V} &= kr^{-p}\hat{r} \\ \nabla \cdot \vec{V} &= krr^{-p}(2-p) = 0\end{aligned}$$

This condition is satisfied for all $r \neq 0$ if and only if $p = 2$. For $p = 2$, the flux is independent of radius, meaning no sources or sinks exist away from the origin. The origin itself is a point source with strength $4\pi k$.

12.

$$\begin{aligned}R &= \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2} \\ \frac{\partial}{\partial x'} \left(\frac{1}{R} \right) &= -\frac{x-x'}{R^3}, \quad \frac{\partial}{\partial y'} \left(\frac{1}{R} \right) = -\frac{y-y'}{R^3}, \quad \frac{\partial}{\partial z'} \left(\frac{1}{R} \right) = -\frac{z-z'}{R^3} \\ \nabla' \left(\frac{1}{R} \right) &= \frac{\vec{r} - \vec{r}'}{R^3} = \frac{\vec{R}}{R^3} = \frac{\hat{R}}{R^2}\end{aligned}$$

The positive sign arises because differentiation is taken with respect to the primed coordinates.